Study Plan for the M.Sc. Degree in Biotechnology

Modified on Spring 24/25 based on Accreditation Requirements & Program Reforms
To be implemented for newly admitted students starting Fall 2025/2026 and onward

First Year

Fall Semester

Course No.	Course Title	Cr. Hrs.
1450502	Biotechnology & Genetic Engineering	3
1450552	Advanced Biochemistry	3
1450xxx	Biotechnology Elective Course	3
1450xxx	Biotechnology Elective Course	3
	Total	**12**

Spring Semester

Course No.	Course Title	Cr. Hrs.
1450512	Advanced Cell Biology	3
1450553	Advanced Molecular Biology	3
1450xxx	Biotechnology Elective Course	3
1450xxx	Biotechnology Elective Course	3
	Total	**12**

Second Year

Course No.	Course Title	Cr. Hrs.
1450501	Research Methodology	1
1450599	Thesis	9
	Total	**10**

Compulsory Courses

Five Biotechnology Compulsory Core Courses (13 Credits):

Course No.	Course Title	Cr. Hrs	Prerequisites
1450501	Research Methodology	1	None
1450502	Biotechnology & Genetic Engineering	3	None
1450512	Advanced Cell Biology	3	None
1450552	Advanced Biochemistry	3	None
1450553	Advanced Molecular Biology	3	None
	Total	**13**	

Elective Courses

Four Biotechnology Elective Courses (12 credits):

Based on the areas of specialization, **four courses Must be** selected from the clusters (**I-III**) listed below:

Cluster I		
Course No.	**Course Title**	**Cr. Hrs.**
1450541	Human Heredity and Molecular Diagnosis of Genetic Diseases	3
1450543	Regulation of Gene Expression	3

Cluster II		
Course No.	**Course Title**	**Cr. Hrs.**
1450545	Advanced Molecular Developmental Biology	3
1450592A	Selected Topics in Biotechnology A	3
1450592B	Selected Topics in Biotechnology B	3

Cluster III		
Course No.	**Course Title**	**Cr. Hrs.**
1450531	Advanced Microbiology	3
1450532	Advanced Immunology	3
1450534	Virology	3